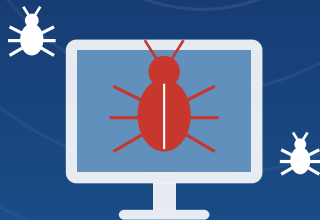




NETWORK SECURITY | MODULE 4

Malicious Software

Lesson 6 – Viruses, Worms and Trojan Horses



Bombs · Trojans · Worms · Viruses · Infection · Prevention · Cure · Hardening



LESSON 6: VIRUSES, WORMS AND TROJAN HORSES

1

Perverse Software

What it is and why it is written

2

Bombs, Trojans, Worms, Viruses

The four kinds and how they differ

3

Evolution and Menace

From 1949 to the Brain virus

4

The Process of Infection

Booting, interrupts and ISRs

5

Classification of Viruses

Boot, system and .COM

6

Famous Viruses and Phases

Friday the 13th, Raindrops and more

7

Prevention and Cure

Precautions and anti-viral programs

8

Hardening and Malicious Code

Guest account, firewalls, six code types



Lesson 6

Viruses, Worms and Trojan Horses



- 1 Define computer viruses and their evolution
- 2 Explain the damage a virus can do
- 3 Understand the process of virus infection
- 4 Identify types of viruses by area of infection
- 5 Describe and apply preventive measures

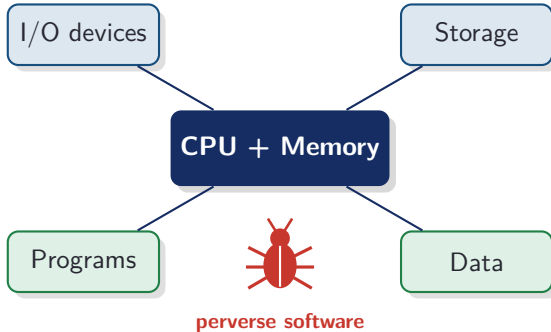
An era of electronic terrorism

The world of IT is going through an era of electronic terrorism in the form of viruses, a problem dangerous enough to **threaten the proper functioning** of computer systems.

Our plan

Evolution → menace → infection →
classification → types → prevention → cure

What Is Perverse Software?



Definition

"Software which causes a **perverse activity**."

Perverse activities include:

- Modifying or completely **destroying data** without the user's intention
- **Unpredictable behaviour** in display, print, etc.
- **Sabotaging** the operating system

A system running perverse software is called an **infected system**.

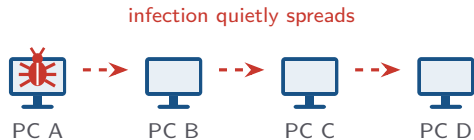
Why Is Perverse Software Written?



The result

All of them aim at producing **disastrous effects**, while the normal user only wants to do something **constructive** and increase productivity.

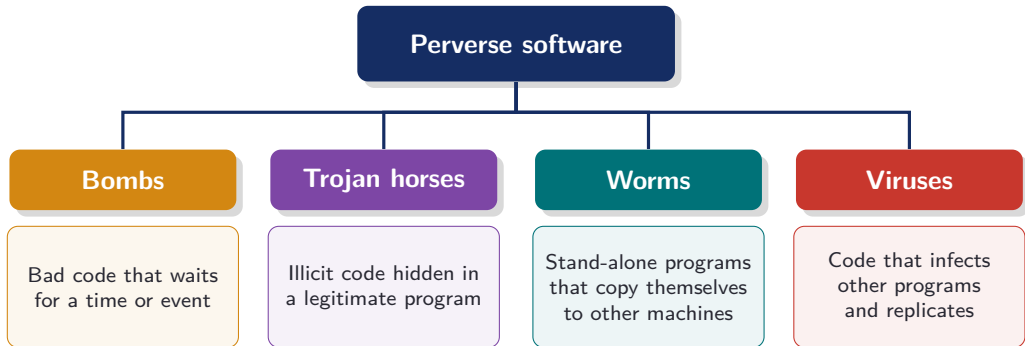
Why Is It So Hard to Notice?



meanwhile every PC **keeps working** as if nothing happened

- Different perverse software causes different activities, but all generate **uncertainty** for users
- Normal MS-DOS operations were designed for **bona fide users**, not to detect such software
- Standard security checks **do not detect** the anomaly
- So an infected system **continues to work**, spreading the infection

Four Kinds of Perverse Software



Note

In everyday speech “virus” is a very broad term that also includes worms and Trojans. In this lesson we keep them separate.



cannot infect other programs,
so a wide epidemic is unlikely

Definition

A piece of **bad code deliberately planted** by an insider or supplier of a program.

- **Triggered** by an event that is logical or time based
- “Explodes” when the conditions are met, causing damage **immediately**
- **Cannot infect other programs**, so it does not propagate
- Two types: **time bombs** and **logic bombs**

Time Bomb vs. Logic Bomb



Time bomb



Named after its physical counterpart: it goes off at a **set date and time**, started by the **computer clock**. It may disrupt the system, modify or destroy stored data.

Logic bomb

```
IF MYFILE is deleted  
THEN overwrite memory  
with zeros
```

Activated by a **certain combination of events**.

In the example, deleting a file called MYFILE destroys the contents of memory.

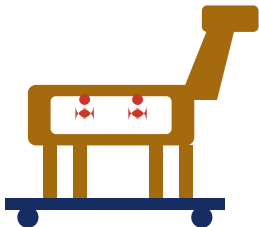
Both types

Can be set to go off at a **future time or event**, and can be planted by someone with inside access.

(b) Trojan Horse



looks like a gift...



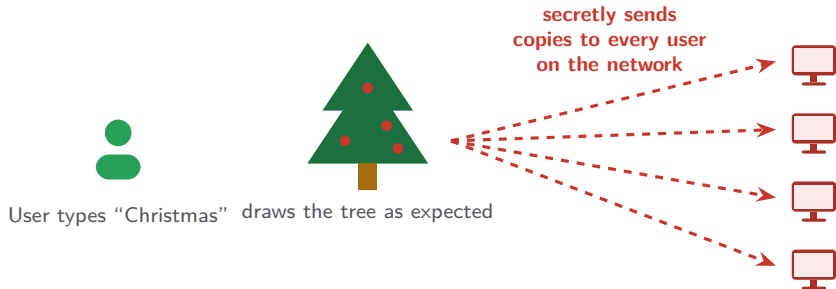
...hides attackers inside

Definition

An **illicit coding** contained in a legitimate program, which causes an illegitimate action. Named after the wooden horse of Troy.

- May change or steal **passwords**, or modify protected records
- Hides in a host and generally does **not damage the host**
- **Cannot copy itself** to other software
- Activated only when the illicit program is **called explicitly**
- Spreads only when an unsuspecting user **copies** the program

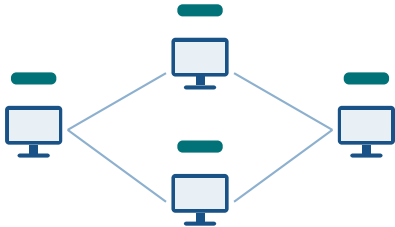
Example: The Christmas Card Trojan



What happened

Detected on IBM's internal e-mail. The program did what it promised, drawing a Christmas tree, but also flooded the network with copies. Other users could not even **save their half-finished work**.

(c) Worms



the worm copies **itself** from machine to machine

- A worm can **relocate itself** and does **not need a host program**
- It copies itself to **another machine** on the network
- Worms are **stand-alone** programs, so they are easier to detect than Trojans and viruses
- Worms store themselves in critical disk areas and have been known to **damage entire LANs**

Worm vs. Trojan

A Trojan needs a host and cannot replicate; a worm needs no host and replicates.



Useful work

Worms can sabotage systems, but can also do **useful tasks**, for example installing a network. Insert a worm and check for it at each node: a node where it never appears can be assumed **not connected**.



not connected



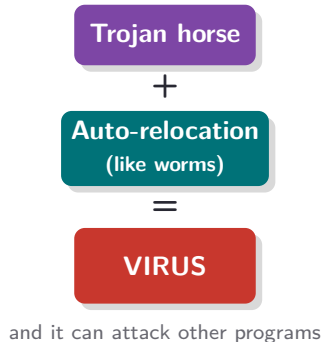
Two named examples

Alarm Clock worm

Places **wake-up calls** on a list of users, travelling through the network to an outgoing terminal

Existential worm

Its sole purpose is to **remain alive**: it only copies itself to several places, causing no damage

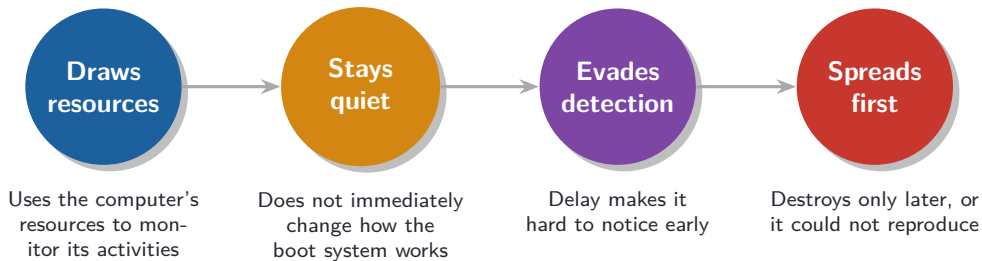


Davis and Gantenbein (1987)

"A Trojan horse program with the capability of **auto-relocation** (same as in worms) and it can **attack other programs**."

- The chronological **successor** of worm programs
- The **most dangerous** perverse software: it reproduces itself
- Attaches to a regularly used program so the host **looks benign**
- Highly **contagious**; can take control of the OS and destroy all data on the disk

A Virus Behaves Like a Parasite



Why the delay?

If a virus **immediately destroyed** its host software, it would never be able to reproduce and spread. So destructive viruses must **delay** their reaction.

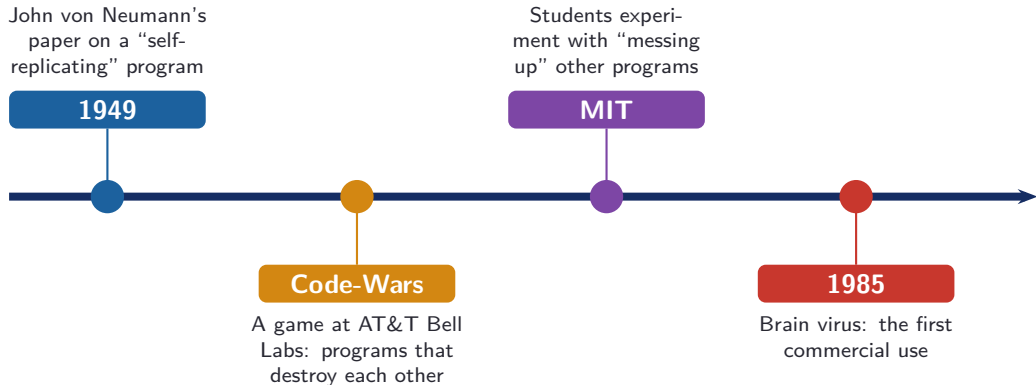
Comparing the Four Kinds (Fig. 6.1)



Perverse software	Requires a host?	Can replicate and relocate?
Bomb	Yes /No	No
Trojan	Yes	No
Worm	No	Yes
Virus	Yes	Yes

Memory trick: a **worm** travels alone; a **virus** needs a host *and* spreads; **bombs** and **Trojans** stay where they are planted.

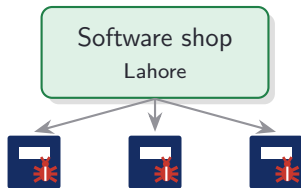
The Evolution of the Virus



From idea to threat

The 1949 idea seemed impossible and was dropped. Code-Wars' authors **kept it secret** because they saw the danger. Harmless hobbies at Bell Labs and MIT gave rise to the **concept of computer viruses**.

1985: The Brain (Pakistani) Virus



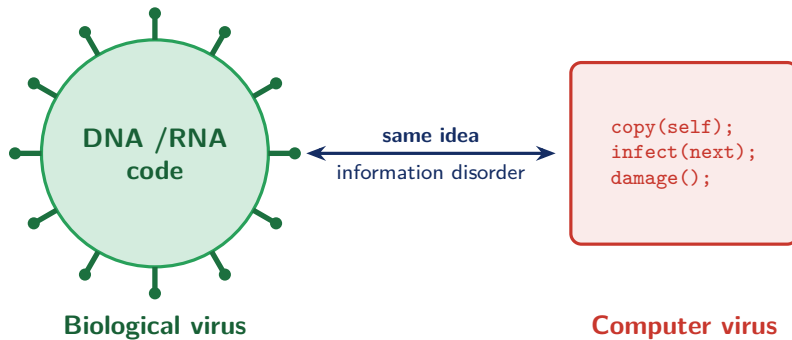
hidden code on every disk
sold to track **software piracy**

- Two Pakistani brothers wanted to **track piracy** of their low-cost software
- A small **self-replicating** snippet was hidden on nearly every disk
- It infected an **unauthorized user's** computer by disrupting operations
- Shows a popup: "**Welcome to the Dungeon**"

Lesson

Such programs multiplied so fast that today they are a threat to every computer.

The Menace: Biological vs. Computer Virus



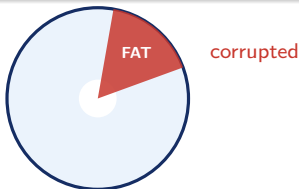
An information disorder

A biological virus takes over a living cell to make thousands of replicas. A computer virus carries **instructional code that makes copies of itself**, and spreads from machine to machine **like a forest fire**.



Data loss

Apart from self-replication, the biggest devastation is **data loss**. Most viruses do this to perform even simple feats.



Two common tricks

- 1 Corrupt the most sensitive disk area: the **File Allocation Table (FAT)** or the directory area
- 2 Modify the system's **interrupt organization**, so every read or write is routed through the virus code, causing interrupt clashes

Even without a “destroy” instruction

A virus can still render a disk full of files **absolutely useless**.

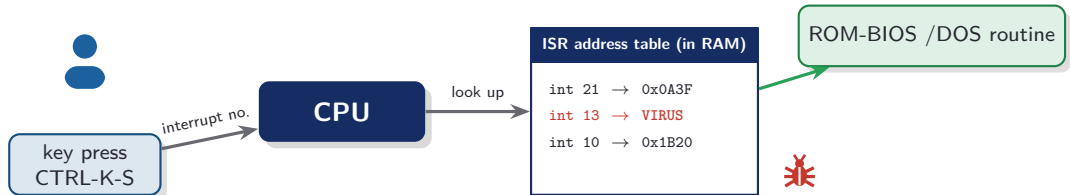
The Process of Infection: Booting



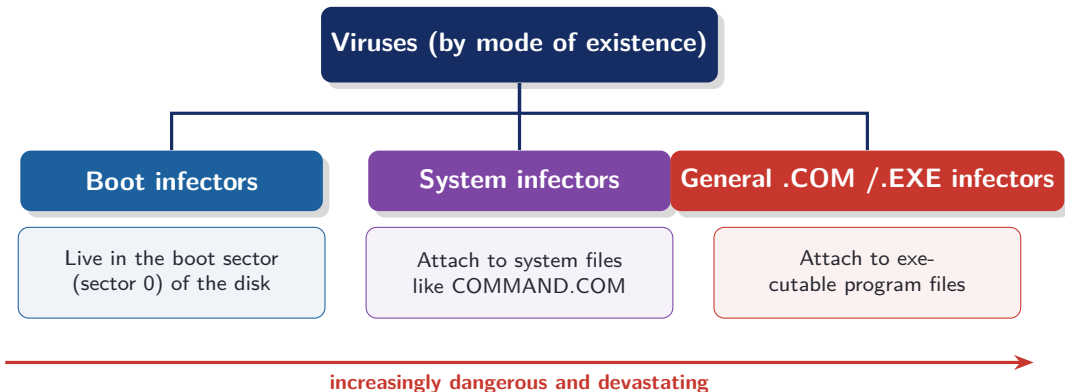
Where infection begins

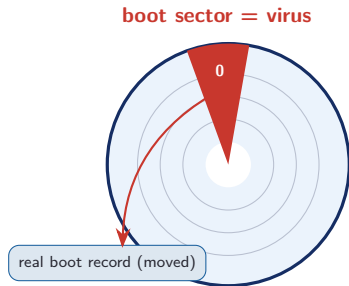
As soon as a computer **boots from a contaminated disk** or **executes an infected program**, whatever viruses are present are activated and begin to spread throughout the system.

The Interrupt Mechanism and ISRs

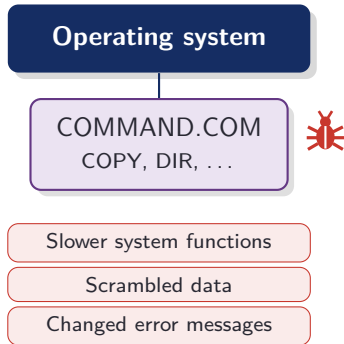


- All I/O on a PC is done by **interrupts**; routines that serve them are **Interrupt Service Routines (ISRs)**
- The CPU looks up the interrupt number in a **table** to find the routine, runs it, and returns control
- The table sits in **RAM**, so user programs can change it, and that is exactly **what a virus does**





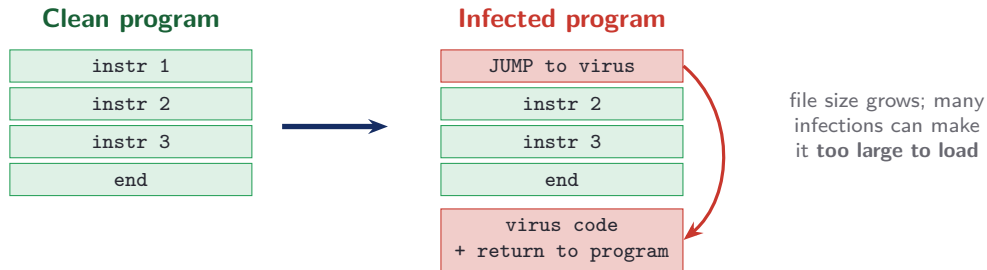
- Physically reside in the **boot sector [0]** of the disk, not in a program file
- Load right after POST and stay in control **at all times**
- Can trap soft booting (CTRL+ALT+DEL) and **infect a clean floppy**
- Move the real boot record elsewhere, and pass control to it after loading, otherwise the disk would not work
- Typically create “**bad sectors**”; stay in memory until shutdown and disk reformat



- Infect the **components of the system itself**
- In MS-DOS, **COMMAND.COM** holds the internal commands (COPY, DIR, ...); without it they cannot load
- Attach to COMMAND.COM or other memory-resident files and **manipulate** them

Differ from boot infectors

They gain control **after** booting, and infect disks that contain the system files only. They may act after some time, or begin subtle changes at once.



- **Most dangerous** of the three classes: they can spread to almost any executable program
- Change the program's instructions into a **jump** to the virus, which runs first and then lets the program continue
- Stay **memory resident** and infect every program loaded after them

The Three Classes Side by Side



Feature	Boot infector	System infector	.COM /.EXE infector
Lives in	Boot sector (0)	COMMAND.COM, system files	Any executable file
Gets control	Right after POST	After booting	When the program runs
Typical sign	Bad sectors	Slow or odd system behaviour	Larger file sizes
Danger	High	High	Highest



Scores

Macintosh. Built-in timer fires 2, 4 and 7 days after infection: print problems, crashes, disk errors. Removal means **deleting all files**.

Brain

One of the first. Pakistani brothers, Lahore. Shows "Welcome to the Dungeon". Destroys data; **highly contagious**.

Lehigh

From Lehigh University. Lives in COMMAND.COM's stack space; spreads via COPY, TYPE, DIR. After **4 copies** it erases all files, boot sector and FAT.



all files deleted

- Attacks COMMAND.COM **and** other executable files
- When the first .COM or .EXE runs, it captures an interrupt and inserts its own code
- In .EXE files the code goes at the **end**: the file grows by **1808 bytes** each time
- In .COM files it goes at the **beginning**
- Programs become too large to load, and later run much **slower**

The worst disaster

If an infected program runs when the system date is **Friday the 13th**, all files get deleted.

Famous Viruses (3): Slug, Raindrops, Happy Birthday



Sunnyvale Slug

```
Greetings from Sunnyvale.  
Can you find me?  
C:\> COPY ...  
file deleted!
```

Modifies COPY so it **deletes** files instead of copying them

Raindrops



Infects .COM files; characters **rain down** the screen with sounds

Happy Birthday 30th

```
Type "Happy Birthday 30th"  
to continue  
Memory: 634 KB
```

Fires on **5 January**; may wipe a 1.2 MB floppy; memory shows **6 KB less**

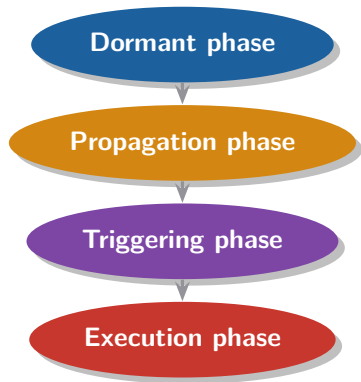
A Rogues' Gallery of Viruses



Marijuana (Stoned)	Joshi	Jerusalem (Friday 13th)	Serum
Zealot	March-6th (Michelangelo)	Frodo (4096)	Desi (Made in India)
Datalock	Gravity	Hong Kong	Liberty
Invader (Anti-CAD)	Black Monday	Tequila	Slow
World Peace	Kanishka	Mummy	Charas
Monkey	Dong_2	Khobar	Green Caterpillar

A comprehensive list cannot be built, because **every other day a new virus** joins it.

The Four Phases of a Virus (Fig. 6.2)

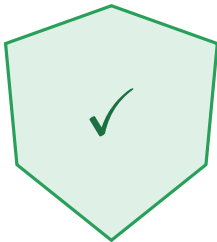


Dormant The virus is **idle**, waiting for a date or event. Not all viruses have this stage.

Propagation The “spread over” phase: the virus places **replicas** of itself; each infected program holds a clone.

Triggering The virus is **activated** to do the task it was created for.

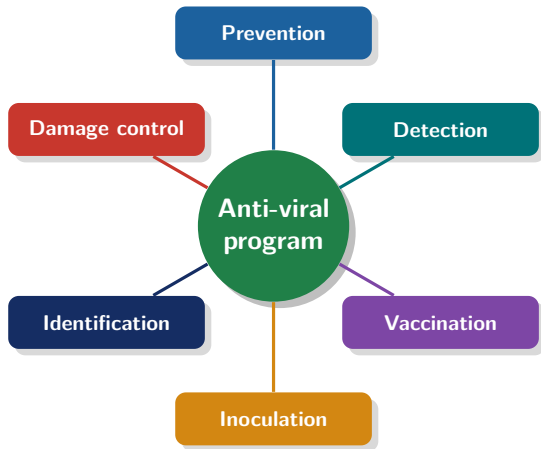
Execution The **actual function** is performed: harmless or damaging.



Viruses are made faster
than the vaccines

Simple precautionary measures

- Add **CHKDSK** to AUTOEXEC.BAT; investigate if hidden files increase
- **Stop** using pirated software
- Use **write-protect tags** on original diskettes
- Keep a proper **backup** of all data and programs
- Copy files carefully: put COPY in a batch file with CHKDSK
- **Reformat** used floppies
- Do not let **unauthorized users** use the system
- **Restrict** outside floppies



Viruses are not omnipotent

They can be cured with anti-viral programs that perform one or more of these six functions.

A good anti-viral utility

Checks whether the system is infected, **stops** viruses from infecting it, and does **not allow modification** of executable files.



Preventers

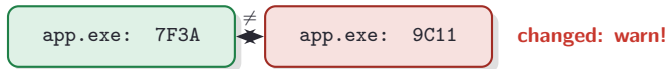
Refuse to let a program stay resident in RAM unless allowed; run only programs on an **approved list**

Detectors

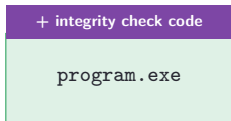
Warn the user **after** a virus is loaded; keep a list of **checksums** of executable files and compare

Identifiers

Rely on the fact that a replicating virus **copies itself**; they search for that copy



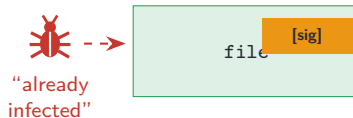
Vaccinators



When run, checks itself and **warns if changed**

Inject some code into executable files. The injected code performs an **integrity check** on the program being executed.

Inoculators



The virus sees its own signature and **goes away**

Insert the **virus signature** into files and areas, so the virus believes they are already infected.



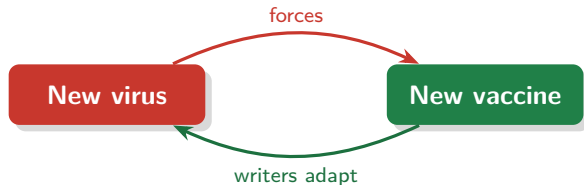
Damage control

Preventive: stop direct access attempts such as formatting and deleting; write-protect the hard disk while testing unfamiliar software.

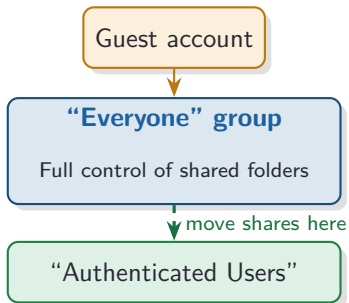
Restorative: keep a copy of the CMOS information, boot sector and FAT in a safe place, like a floppy.

No 100% foolproof anti-virus

- A virus can hide in many ways
- Virus writers keep **altering** their code to avoid detection
- So there is no perfect anti-virus, and in principle **there never will be**



an endless race: keep up to date



- In Windows 2000 the Guest account is **disabled by default**; enabling it allows **anonymous** access
- A shared folder's default permission is: **Everyone has full control**
- Guest is part of "Everyone", so security is **compromised**

Standard practice

Remove share permissions from "Everyone" and give them to "Authenticated Users".

Steps to Configure the Guest Account



1

Log in as **Administrator**

2

Start → Programs → Administrative Tools → **Computer Management**

3

Open **Local Users and Groups**, then the **Users** folder

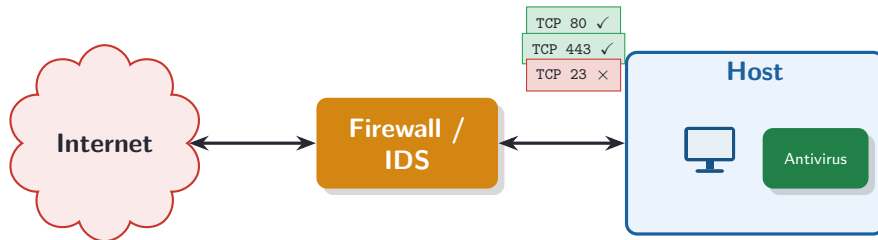
4

Right-click the Guest account and **rename** it to a non-obvious name

5

Right-click the renamed account and choose **Set Password**

A renamed, password-protected account gives attackers **nothing obvious to guess**.



Next step after accounts

Install a **host-based antivirus** and a **firewall/intrusion detection system**. Configure which **TCP/UDP ports** can be accessed, so undesired communication does not happen on any port.

Six Types of Malicious Code



Viruses

Inserted into other executable code; runs when it runs

Worms

Run independently and travel across network connections

Trojan horses

Look innocent and useful but hide a harmful function

Back doors /trap doors

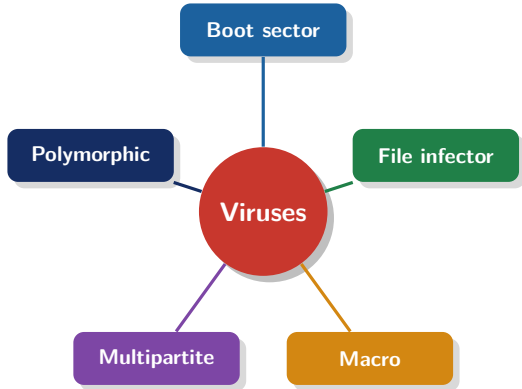
Secret code that bypasses normal authentication

Logic bombs

Lie dormant in software until a condition is met

Bacteria /rabbits

Replicate to eat all CPU, memory or disk space



A true virus is code **inserted into other executable code**, so it runs whenever the program runs.

Boot sector Infect the DOS boot sector or master boot record; run at boot

File infectors Attach to executables; run with the program

Macro Ride inside documents with macros; run when opened

Multipartite Combine boot sector + file infector

Polymorphic Change themselves on replication to dodge **signature** scanning



Worm

Executes independently and travels between machines. Self-copies spread **like a flood**, slowing or breaking network services.

Trojan horse

Appears innocent and useful but hides an **unintended function**. Does not replicate, but can steal passwords, delete data or format disks.

Back door /trap door

Code that grants **special access**, bypassing authentication. Debugging shortcuts left in released software are a risk.



Logic bombs

Programmed threats that lie **dormant** in commonly used software for a long time until **pre-conditions** are met, such as a particular day. They come embedded in some programs.



Bacteria /rabbits

Do **not damage files**. Their aim is to **deny access** to resources by replicating until they consume all processor capacity, memory or disk space.





Damage caused

Ranges from **merely annoying** to **catastrophic**: loss of data or services, disclosure of information. A software firm that ships malicious code risks its **reputation** and legal trouble.

Who writes virus code?

-  **Disgruntled employees**
-  **Spies**
-  **Experimenters**
-  **Publicity hounds**
-  **Political activists**

Protecting your system

- Be careful about installing **new software**
- Never install binaries from **untrustworthy sources**
- Test new software first on a **non-critical system**



What malicious software is

- Perverse software and its motives
- Bombs, Trojans, worms, viruses (host? replicate?)
- Evolution: 1949 to Brain (1985)
- Booting, interrupts and ISRs
- Boot, system and .COM/.EXE infectors
- Famous viruses; four phases

How we fight it

- Precautions: prevention is better than cure
- Six anti-viral functions
- Damage control; no perfect anti-virus
- Hardening: Guest account, firewall, ports
- Six types of malicious code

Review questions

1. Define perverse software.
2. Explain the classification of viruses.
3. Explain the different types of virus.
4. Write a short note on hardening network activity.



- ① William Stallings, *Cryptography and Network Security*, PHI Publishers.
- ② Charlie Kaufman, Radia Perlman, *Network Security*, Pearson Education.
- ③ Bruce Schneier, *Applied Cryptography*, John Wiley & Sons.
- ④ Roberta Bragg, *Network Security: The Complete Reference*, TMH.

Thank You! Questions?